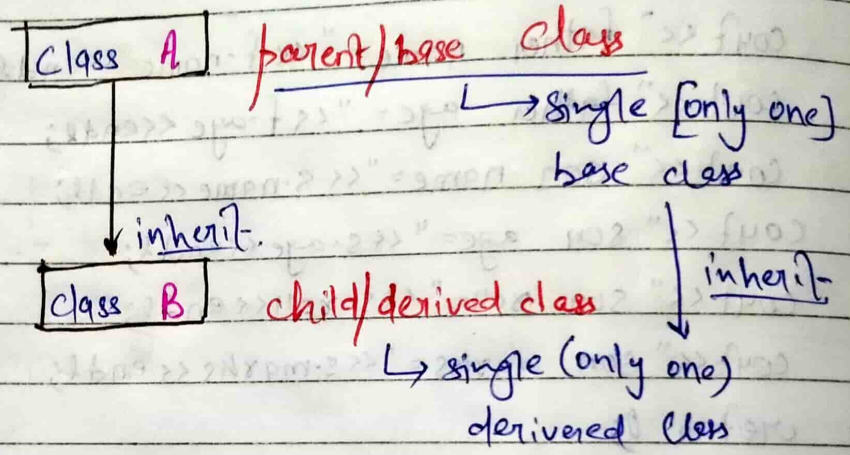
→ Types of Inheritance :-

(i) Single Inheritance :

A single derived class inheritance from a single base class.



→ // C++ program;

```
#include<iostream>
using namespace std;
class parent   // single base class
{
    public :
        string name;
};

class child // single derived class
{
```

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```
class child : public parent   // single derived class
{
    public :
      int roll;
      int marks;
};

int main()
{
    parents p;
    child c;
    p.name= "vasudev";
    c.name= "krishna";   // inherited from parents
    c.roll= 123100;
    c.marks= 100;

    cout<<" parent name= " << p.name<<endl;
    cout<<" child name= " <<c.name<<endl;
    cout<<" child roll= " <<c.roll<<endl;
    cout<<" child marks= " <<c.marks;
  return 0;
}
```

(ii) Multiple Inheritance :

A derived class inherits from more than one base class.

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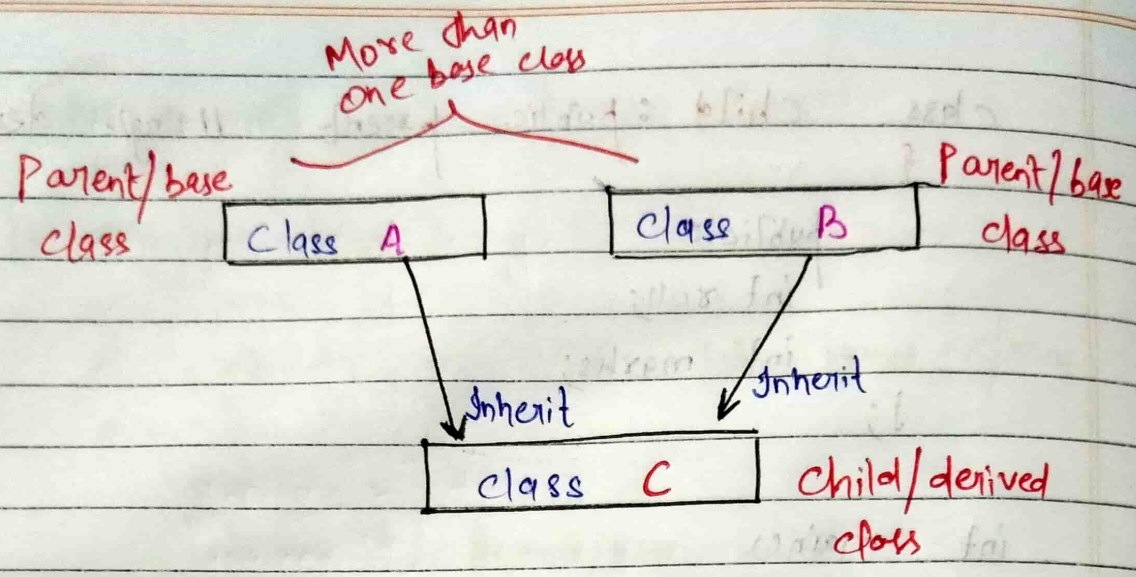

~~// C++ program~~

→ Syntax :-

```
class A
{
    // body
};

class B
{
    // body

};
```

~~class C: public~~

```
class C: <access specifier> A, <access specifier> B
{
    // body
};
```

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→ // C++ program

```
#include<iostream>
using namespace std;
Class grand_parents   // base class → Grandparents
{
    public :
        string name;
        int age;
        int dob;
};

Class parents   // base class → parents
{
    public :
        int salary;
};

// Derived class → child (inherits from both
// grand_parents and parents)

~~Class child : public grandpar~~

Class child : public grand_parents, public parents
{
    public :
        string post;
};
```

Akash

```
int main()
{
    grand_parents g;
    parents p;
    child c;

    g.name = "dasrath";
    g.age = 99;
    g.dob = 23;
    p.salary = 50000;
    c.name = "luv";   // inherit from grand_parent
    c.age = 5;        //  "      "      "
    c.dob = 20;       //  "      "      "
    c.salary = 150000 // inherit from parent
    c.post = "archer";

    cout<<"grandparent name=" <<g.name <<endl;
    cout<<"grandparent age=" << g.age <<endl;
    cout<<"grandparent date of birth <<g.dob<<endl;
    cout<<" parent salary" << p.salary << endl;
    cout<<" child name=" << c.name<<endl;
    cout<<" child age=" << c.age<<endl;
    cout<<" child date of birth = " << c.dob<<endl;
    cout<<" child salary = " << c.salary <<endl;
    cout<<" child post = " <<c.post <<endl;
    return 0;
}
```

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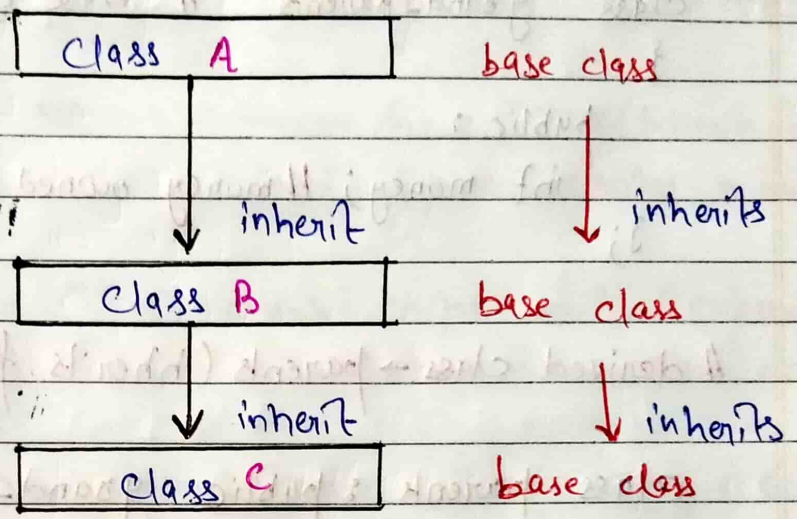
(iii) Multilevel inheritance :-

A derived class inherits from another derived class.



→ Syntax :-

```
Class A
{
    // body
};

class B : <access specifier> A
{
    // body

};
class C : <access specifier> B
{
    // body
};
```

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→ // C++ program

```
#include<iostream>
using namespace std;
class grand_parents   // base class → grand_parent
{
    public :
        int money; // money owned by the grandparents
};

// derived class → parents (inherits from grandparents)

class parents : public grand_parents
{
    public :
        string business; // business started by the parents
};

// derived class → child (inherits from parents)

class child : public parents
{
    public :
        string company; // company established by child
};

int main()
{
```

Akash

```
grand-parents g;
parents p;
child c;

g.money = 100000; // grand parents have ₹100000

p.money = 50000; // parents have ₹50000 (inherits from grand parents)
p.business = "soft drink"; // parent started a business

c.business = "soft drink + chips + ...."; // expanding business
c.company = "coca cola"; // child start their own company

cout<<" grand parent money = "<<g.money <<endl;
cout<<" parent money= " << p.money<<endl;
cout<<" parents business= "<< p.business <<endl;
cout<<" child business= "<< c.business <<endl;
cout<<" child company = " << c.company <<endl;

return 0;
}
```

(iv) Hierarchical inheritance :

One base class is inherited by multiple derived classes.

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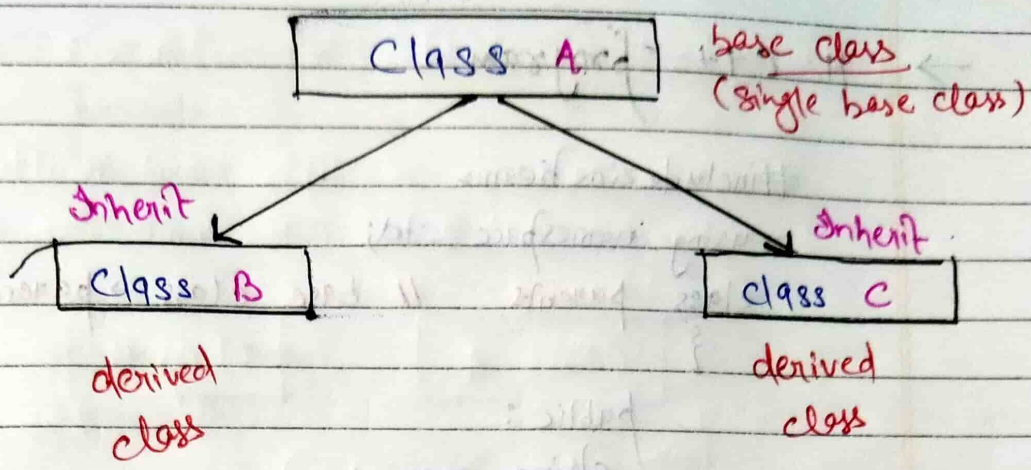

→ Syntax;

```
Class A
{
    // body
};

Class B :< access specifier> A
{
    // body
};

Class C : < access specifier> A
{
    // body
};
```

Akash

→ // C++ program

```
#include<iostream>
using namespace std;
class parents    // base class → parents
{
    public:
        string name;
        int age;
};

// derived class → child_1 (inherits from parents)

class child_1 : public parents
{
    public:
        string subject;
};

// derived class → child_2 (inherits from parents)

class child_2 : public parents
{
    public:
        string sports;
};

int main()
```

Akash

```
{
    parents p;
    child_1 c1;
    child_2 c2;

    p.name = "ram";
    p.age = 30;

    c1.name = "Luv";
    c1.age = 4;
    c1.subject = "mathematics";

    c2.name = "kush";
    c2.age = 5;
    c2.sports = "archer";

    cout<<"parents name = "<<p.name<<endl;
    cout<<" parents age = "<<p.age<<endl;
    cout<<" child 1 name = "<<c1.name<<endl;
    cout<<" child 1 age = "<<c1.age<<endl;
    cout<<" child 1 subject = "<<c1.subject<<endl;
    cout<<" child 2 name = "<<c2.name<<endl;
    cout<<" child 2 age = "<<c2.age<<endl;
    cout<<" child 2 sports = "<<c2.sports<<endl;

    return 0;
}
```

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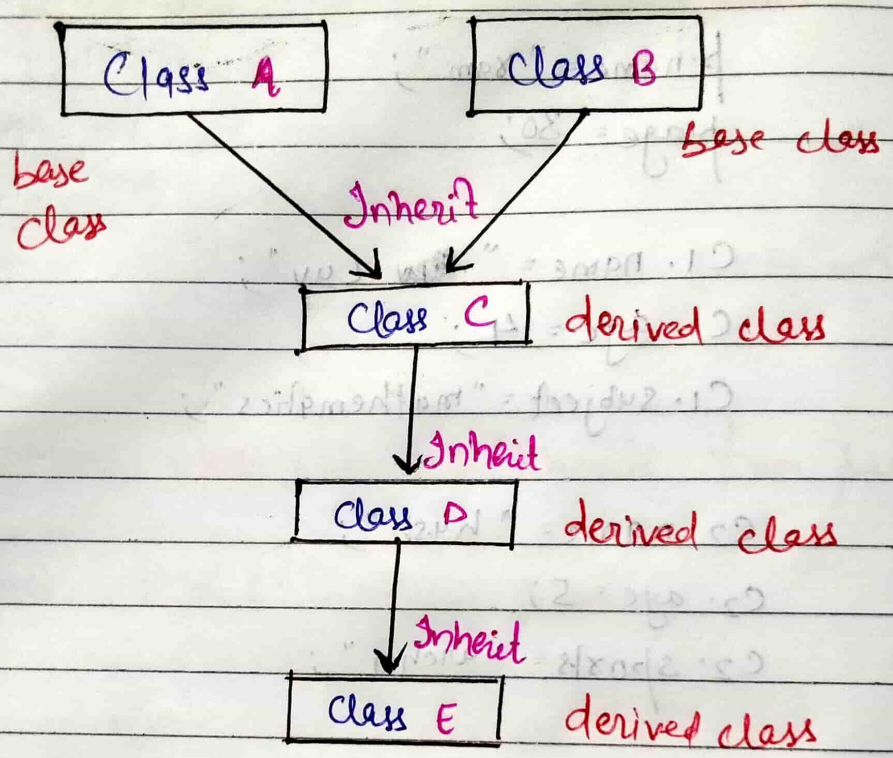
(V) Hybrid (Virtual) inheritance :

Combination of multiple and multilevel inheritance. and it is combination of hierarchical also.



→ C++ program

```
#include <iostream>
using namespace std;
class base1
{
    public:
        String name;
};

class base2
{
```

Akash

```
    public:
      int age;
};

class derived1 : public base1, public base2
    {
        public:
          int roll;
    };

class derived2 : public derived 1
    {
        public:
          int marks;
    };

class derived3 : public derived2
    {
        public:
          int cgpa;
    };

int main()
  {
      base1 b1;
      base2 b2;
      derived1 d1;
      derived2 d2;
      derived3 d3;
```

Akash

```
b1.name= "akash kumar";
b2.age=20;
d1.name="akka";
d1.age="200";
d1.roll=12355;
d2.roll=123100;
d2.marks=121;
d3.marks=99;
d3.cgpa=2;

cout<<"-----";
 - - - -
 - - -
 - -
 -
 .

return 0;

}
```

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→ Access specifier in inheritance :-

Access specifier	Public Inheritance	Protected Inheritance	Private Inheritance
Public member	Public	protected	private
Protected member	Protected	Protected	private
Private member	Not Inherited	Not Inherited	Not Inherited

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